Assignment number 1 (intro to programming)

1. Ask the user for a string that contains both letters and numbers (e.g., "Py2th0on3"). Write a program that separates them into two lists: one for the letters and one for the numbers. Calculate the sum of the numbers found and join the letters to form a word.

- Example Input: "Py2th0on3"
- Example Output: Word: "Python", Sum: 5

2. Ask a user to enter a password. Check if it is valid based on these rules: Length must be between 8 and 16 characters, and it must contain at least one uppercase letter and one special character (e.g., @, #, $).

- Example Input: "Guest@123"
- Example Output: "Password is Secure."

3. A store gives discounts based on the total price: Under $50 (No discount), $50 to $100 (10% discount), and Over $100 (20% discount). If the user is a "Member," they get an additional 5% off the final calculated price.

- Example Input: Price: 100, Member: Yes
- Example Output: Final Total: $85.5 (Note: 100 - 10% = 90; then 90 - 5% = 85.5)

4. Ask the user to input two separate lists of names (comma-separated). Create a third list that contains only the names that appear in both lists.

- Example Input: List 1: "Ali, Ben, Sara", List 2: "Ben, Joy, Ali"
- Example Output: Common names: ['Ali', 'Ben']

5. Create a dictionary where keys are student names and values are lists of 3 test scores. Write a program that calculates the average for each student. If the average is > 90, print the student's name followed by "Honor Roll."

- Example Input: {'Sam': [95, 90, 92]}
- Example Output: Sam: Average 92.3 - Honor Roll

6. Ask the user for two words of the same length. Create a new string by interleaving their characters (one from the first word, one from the second).

- Example Input: word1 = "abc", word2 = "xyz"
- Example Output: "axbycz"

7. Ask for a multi-digit number. Sum its digits. If the result is more than one digit, keep summing the digits until you get a single-digit result.

- Example Input: 98
- Example Output: Digital Root: 8 (9+8=17, 1+7=8)

8. Given a list of numbers with many duplicates, return a list of tuples showing each unique element and its frequency (how many times it appeared).

- Example Input: [1, 1, 2, 3, 3, 3, 4]
- Example Output: [(1, 2), (2, 1), (3, 3), (4, 1)]

9. Ask the user for a number n. Calculate the factorial (n!) using a loop, and then check if the resulting factorial is divisible by the sum of numbers from 1 to n.

- Example Input: 4
- Example Output: Factorial is 24, Sum is 10. 24 is not divisible by 10.

10.Start with a dictionary of items and prices: {'apple': 0.5, 'bread': 2.0, 'milk': 1.5}. Allow the user to "Add" or "Remove" items from a cart. If they add an item not in the store, print an error. Finally, print the total bill.

- Example Input: Add 'apple', Add 'milk'
- Example Output: Total Bill: $2.0

11.Check if a full sentence is a palindrome, ignoring spaces and capitalization.

- Example Input: "Race car"
- Example Output: True

12.Create a "square" of numbers based on user input n. Each row should contain the row number repeated n times.

- Example Input: n=3
- Example Output:

  1 1 1

  2 2 2

  3 3 3

13.In a list of numbers, find all "peaks." A peak is a number that is greater than the number to its left and the number to its right.

- Example Input: [1, 5, 2, 6, 3]
- Example Output: Peaks: [5, 6]

14.Take a dictionary like {'A': 1, 'B': 2, 'C': 1} and flip it. If two keys have the same value, the new value should be a list of those keys.

- Example Input: {'A': 1, 'B': 2, 'C': 1}
- Example Output: {1: ['A', 'C'], 2: ['B']}

15. Create a dictionary where the keys are the vowels found in a user's string and the values are lists of the indices where those vowels appear.

- Example Input: "banana"
- Example Output: {'a': [1, 3, 5]}